

Lineage: Document-Level Parent Resolution for Partially Specified Hierarchical Mentions

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Abstract

Hierarchical entity mentions in long-form documents are often only partially specified: a document may refer to a battalion, company, or platoon without restating the parent unit to which it belongs. We introduce LINEAGE, a document-level parent-resolution task for partially specified hierarchical mentions in parent-scoped homonymous nomenclatures. In such systems, local child designators such as *1st Battalion* or *Company A* are systematically reused under different parents, so the mention surface is not globally identifying. Given a target mention, document context, and a candidate parent pool, a system must select the document-licensed parent or abstain when the context is insufficient or ambiguous. We construct a focused expert-labeled benchmark from unresolved low-echelon references in a frozen historical information-extraction resource. Gold v1 contains 408 strict-deduplicated rows, with 333 instances used for main evaluation and 75 diagnostic or excluded cases. We implement LINEAGE with a Scope-aware Constrained Parent Resolver that combines high-recall candidate generation, hierarchy constraints, scope licensing, contextual ranking, and abstention. On the Gold v1 test split, forced rankers reach higher resolved-only accuracy, but the scope-aware resolver improves abstention-aware macro-F1 from 0.250 to 0.517 while increasing selective accuracy to 0.601. These results support evaluating parent resolution as document-level scope licensing with abstention rather than as forced parent ranking alone.

1 Introduction

Entity mentions in hierarchically organized domains are often locally named rather than globally named. A document may introduce a parent orga-

nization and later refer only to a child designator, such as *1st Battalion*, *Company A*, or *3d Platoon*. These strings are not unique identifiers: the same local designator can recur under multiple regiments, battalions, divisions, agencies, departments, or legal sections. Resolving such mentions requires recovering the omitted parent that is licensed by the surrounding document context.

We call this problem *document-level parent resolution for partially specified hierarchical mentions*. The title term LINEAGE refers to the omitted parent line that a resolver must recover; it is not a domain-specific acronym. The task is related to entity linking, bridging resolution, and document-level relation extraction, but differs from each: the system selects an omitted parent argument and must abstain when the document does not license a unique attachment (Kolitsas et al., 2018; Kobayashi and Ng, 2020; Yao et al., 2019).

The central observation is that parent resolution is a *scope-licensing* problem. A parent is plausible only if it has been introduced, remains active, and is compatible with the hierarchy around the target mention. Globally frequent or salient parents can be wrong if they do not license the local designator at the target position. We therefore augment contextual ranking with parent-introduction priors, discourse scope memory, homonymy-aware features, and explicit abstention.

We make three contributions. First, we define document-level parent resolution for parent-scoped homonymous hierarchical nomenclatures. Second, we introduce LINEAGE, a focused expert-labeled benchmark with explicit abstention and resource-gap diagnostics. Third, we evaluate forced-ranking and scope-aware models, showing that abstention-aware evaluation captures behavior missed by resolved-only Acc@1.

2 Task Definition

Parent-scoped homonymous nomenclatures.

We target hierarchical naming systems in which local child designators are systematically reused under different parents. A child mention is therefore not globally identifiable from its surface form alone. This excludes arbitrary taxonomy completion or generic hypernym discovery: the target is a document-local parent attachment licensed by discourse scope (Camacho-Collados et al., 2018).

Local designators and cores. We distinguish two representations. The *local designator* preserves the typed child expression, such as *1st Battalion*, *Company A*, or a transliterated “1st Battalion” form (“1st Battalion”). The *designator core* removes the type/echelon and keeps the ordinal or letter, such as *1st* or *A*. Homonymy is computed over a level-conditioned key:

$$h(\ell, z) = |\{p \in \mathcal{P} : (\ell, z, p) \text{ is possible}\}|, \quad (1)$$

where ℓ is the child level and z is the designator core. This avoids collapsing *1st Battalion*, *1st Company*, and *1st Platoon* into one class.

Output. For each target mention m , the system receives a document D and candidate parent pool $C(m)$, then outputs either a parent $c \in C(m)$ or an abstention decision. Gold v1 uses seven annotation outcomes, but the main evaluation contains only RESOLVEDPARENT, ABSTAININSUFFICIENTCONTEXT, and ABSTAINAMBIGUOUS. RESOURCEGAP and GENERICPARENTONLY are retained as candidate-construction diagnostics; extraction artifacts and same-surface span-ambiguous cases are excluded. Model-internal ABSTAINNOSCOPE and ABSTAINNOEVIDENCE outputs map to ABSTAININSUFFICIENTCONTEXT for main evaluation.

3 Dataset

LINEAGE is derived from unresolved low-echelon references preserved in a frozen historical information-extraction package. The source resource preserves mention-level provenance through `mention_id`, `source_id`, and `evidence_id`, and avoids forcing bare low-echelon mentions

Category	Count
Strict-deduplicated rows	408
Main evaluation rows	333
Development / test	84 / 249
Candidate rows	3,996
Candidates per instance	12
RESOLVEDPARENT with one gold cand.	199/199
RESOURCEGAP diagnostics	58
Other diagnostic/excluded rows	17

Table 1: Gold v1 and model-ready candidate statistics. Main evaluation uses RESOLVEDPARENT, ABSTAININSUFFICIENTCONTEXT, and ABSTAINAMBIGUOUS only.

into unstable parent chains. Target mentions satisfy `map_status=unresolved`, `mapping_note=bare low-echelon` coreference without stable parent, and a child echelon in `{platoon, company, battalion}`.

Gold v1 contains 500 raw provenance rows. Strict deduplication by `source_id @@ evidence_id @@ raw_mention` yields 408 rows. The main evaluation set contains 333 instances, split into 84 development and 249 test instances. Gold is not used for training; development data is used for threshold tuning and calibration, and test data only for final evaluation. The model-ready candidate table contains 3,996 rows, exactly 12 candidates per instance, and all 199 RESOLVEDPARENT instances have exactly one gold candidate.

We manually curated 60 apparent resource-gap cases: 42 parents existed in the original lexicon but were missed by candidate generation, 16 required new canonical-unit candidates, and 2 were reclassified as insufficient-context abstentions. The final strict-deduplicated diagnostic set therefore contains 58 RESOURCEGAP instances. We report these as candidate/resource diagnostics, not ranking failures.

Gold labels are not used for training. We mined 237 explicit hierarchy-chain matches and retained 101 after leakage filtering. The silver data is explicit-chain-only and used as auxiliary evidence; the main evaluation concerns partially specified document-level mentions.

The annotation process yielded a natural-overlap reliability audit, not a balanced double-annotation design. After collapsing same-annotator repeti-

tions, the audit contained 15 groups and 36 annotator votes. Pre-adjudication exact agreement on the final decision–parent tuple was 13/15.

4 Method

We implement LINEAGE with a Scope-aware Constrained Parent Resolver. The resolver uses local designator extraction, homonymy degree estimation, high-recall candidate generation, hierarchy and ontology constraints, a parent-introduction prior, discourse scope memory, contextual candidate ranking, and scope-aware abstention.

Scope-licensing prior. Let t_m be the target position and $T(c)$ the document positions where candidate c appears. We define nearest previous and following distances:

$$b(c, m) = \min_{t \in T(c), t < t_m} (t_m - t), \quad (2)$$

$$a(c, m) = \min_{t \in T(c), t > t_m} (t - t_m). \quad (3)$$

If no such position exists, the distance is infinite. Recency is $R(c, m) = \exp(-b(c, m)/\tau_\ell)$ when $b(c, m)$ is finite and 0 otherwise. We use E_{pre} and E_{post} for explicit hierarchy chains where the parent appears before the child or in same-sentence apposition after the child.

$$\begin{aligned} I(c, m) = & \alpha_{\text{pre}} E_{\text{pre}}(c, m) \\ & + \alpha_{\text{post}}^\ell E_{\text{post}}(c, m) \\ & + \alpha_{\text{rec}} R(c, m) \\ & - \alpha_{\text{late}}^\ell L(c, m) \\ & - \alpha_{\text{stale}} S_{\text{stale}}(c, m). \end{aligned} \quad (4)$$

where $L(c, m)$ penalizes candidates first introduced after the target outside same-sentence apposition, and S_{stale} penalizes stale prior scopes. Korean official histories strongly prefer preposed parent scopes, while English military prose often permits postposed appositive chains. We therefore use source-specific directionality priors. Homonymy scaling is $\omega(m) = 1 + \eta \log(1 + h(\ell(m), z(m)))$, and $I'(c, m) = \omega(m)I(c, m)$.

Scope memory and ranking. The resolver maintains a level-specific active parent scope. Scope memory is

$$G(c, m) = \mathbb{K}[c = \text{Scope}_{\text{level}(m)}(t_m)] \cdot M(c, m), \quad (5)$$

where $M(c, m)$ decays with distance, section-boundary crossings, and intervening competing parents. The final score reserves E_{chain} for explicit hierarchy-chain evidence and places event/place/time overlap inside F_θ :

$$\begin{aligned} S(c \mid m, D) = & F_\theta(m, c, D) \\ & + \lambda_H H(c, m) + \lambda_O O(c, m) \\ & + \lambda_I I'(c, m) + \lambda_G G(c, m) \\ & + \lambda_E E_{\text{chain}}(c, m). \end{aligned} \quad (6)$$

Abstention. The resolver abstains when no candidate receives sufficient scope or score evidence. ABSTAINNOSCOPE and ABSTAINNOEVIDENCE are model-internal diagnostics mapped to ABSTAININSUFFICIENTCONTEXT; RESOURCEGAP is reported separately.

5 Experiments

Candidate generation. The frozen candidate pool covers all 333 main-evaluation instances. On test RESOLVEDPARENT cases, it obtains Recall@1=0.658, Recall@3=0.886, Recall@5=0.946, and Recall@10=1.000. Thus, the gold parent is available by $k = 10$, but top-rank selection and abstention remain non-trivial.

Results. Table 2 shows that forced parent ranking is incomplete for this task. M2, a forced variant with a parent-introduction prior, obtains the highest resolved-only Acc@1 among evaluated variants (0.765), but its abstention-aware macro-F1 remains 0.250 because it always predicts a parent. The proposed M5 scope-aware abstention model lowers coverage to 0.635 and resolved-only Acc@1 to 0.638, but improves macro-F1 to 0.517 and selective accuracy to 0.601. This supports the central evaluation claim: document-level parent resolution should not be evaluated as forced parent ranking alone.

Ablation and diagnostics. The clearest ablation effect is scope-aware abstention: removing it increases resolved-only Acc@1 by 0.107 because the model resolves more often, but decreases macro-F1 by 0.267. Hierarchy constraints also matter for valid parent ranking. Other scope-prior components show smaller aggregate effects in this focused

Model	Acc@1	Macro-F1	MRR	Coverage	SelAcc
B2 nearest previous	0.664	0.250	0.771	1.000	0.398
B3 same-row/window	0.611	0.489	0.775	0.743	0.492
B4 BM25 ranker	0.591	0.250	0.731	1.000	0.353
B5 scope feature ranker	0.745	0.250	0.843	1.000	0.446
M2 parent prior, forced	0.765	0.250	0.859	1.000	0.458
M5 scope-aware abstention	0.638	0.517	0.845	0.635	0.601

Table 2: Gold v1 test results. Forced rankers achieve higher resolved-only Acc@1 but do not model abstention. M5 lowers coverage through scope-aware abstention and improves abstention-aware macro-F1 and selective accuracy.

Ablation	Δ Acc@1	Δ F1
w/o abstention	+0.107	-0.267
w/o hierarchy constraints	-0.107	-0.022
w/o ontology constraints	-0.020	+0.002
w/o scope memory	-0.013	+0.009
w/o parent prior	0.000	0.000
w/o homonymy scaling	0.000	0.000
shared source prior	0.000	0.000

Table 3: Test ablations relative to M5. Removing abstention raises forced Acc@1 but sharply reduces task-level macro-F1.

benchmark; we interpret them as diagnostically useful rather than independently large performance drivers. Diagnostics show that same-evidence support is comparatively easy (Acc@1=0.893), whereas longer-distance support is much harder: the 4–10 bucket has Acc@1=0.083 and the 11–25 bucket has Acc@1=0.000.

6 Related Work

Bridging resolution links an expression to a non-identical but associated antecedent (Kobayashi and Ng, 2020; Hou, 2020). LINEAGE is narrower: the relation is a typed parent attachment under hierarchy constraints. Entity linking provides candidate generation and ranking machinery (Kolitsas et al., 2018), but LINEAGE selects an omitted parent rather than the mention’s direct referent. Hypernym discovery retrieves lexical hypernyms (Camacho-Collados et al., 2018); our parent is document-local and scope-licensed. Document-level relation extraction requires multi-sentence reasoning (Yao et al., 2019), but typically assumes both relation arguments are explicit.

7 Conclusion

We introduced LINEAGE, a document-level parent-resolution task for partially specified hierarchical mentions in parent-scoped homonymous nomenclatures. Gold v1 provides a focused expert-labeled benchmark with explicit abstention and resource-gap diagnostics. Experiments show that forced rankers can perform well on resolved-only Acc@1 but fail the abstention-aware requirement of the task. A scope-aware abstention resolver better matches the task definition by reducing unsupported parent attachments and improving macro-F1 and selective accuracy.

Limitations

The benchmark is focused rather than large-scale, and its current case domain is historical military narrative. The task framing is domain-general, but future work should test other parent-scoped nomenclatures. The development split is small, so threshold tuning and ablation interpretation should be treated cautiously. Gold v1 does not include character offsets for repeated same-surface mentions within a single evidence row; such cases are separated as EXCLUDESPANAMBIGUOUS. The silver data is explicit-chain-only and auxiliary. The final resolver is feature-based; we did not run a large neural reranker. Some source-position and paragraph/sentence features are proxy-derived from the frozen evidence backbone. Finally, the ablation results show strong support for abstention and hierarchy constraints, but weaker aggregate evidence for individual scope-prior components.

Ethical Considerations

The source domain contains historical military narratives. The dataset is intended for retrospective document analysis and information extraction, not for contemporary operational intelligence or decision support. We separate resource gaps from ranking errors to avoid overclaiming system capability when the candidate universe is incomplete. Any public release should respect source-text redistribution constraints and, where necessary, release annotations, candidate metadata, and reconstruction scripts rather than unrestricted copies of source text.

References

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